

compatible across various DVD drives and players than DVD-R is. DVD+RW is a rewriteable version of DVD+R, and both these have the same capacity—4.37 GB. These formats have been strongly backed by Sony and HP.

Then there are DVD-R DL and DVD+R DL, the dual-layer formats, which have capacities of 8.5 GB. They are nothing but DVD-R and DVD+R with two layers to store data. They require drives that support these formats, which are quite common today.

Finally, DVD-RAM is a format that allows repeated recording and erasing, but it enjoys the lowest compatibility amongst all the DVD formats—it is supported by only a select few drives. DVD-RAM discs are typically housed in cartridges. This format supports 4.7 GB, which is the highest capacity in the DVD realm. DVD-RAM discs find application mainly in real-time video capture.

In addition to this—and we hope we're not adding to the confusion—each of these media has a double-sided version, which you can think of as two discs with the label sides stuck together—you can flip them over and write on the other side. Naturally, they have twice the capacity of the single-sided media, but they're hard to find.



INTERNAL DVD-WRITERS

These sport the IDE interface, and connect to the PC using an IDE cable. They are more popular than their external counterparts because of their considerably lower prices and due to them being faster because, well, the interface is faster. Other than this, there isn't much of a difference between an internal and an external drive. Regular home and office PC users and office PC users generally opt for internal drives.

We tested twelve internal DVD-Writers in this shootout. These were the Asus DRW-1608P2S, BenQ DW1640 and DW1625, Gigabyte GO-1623A-RH, LG GSA-4167B, Lite-On SHW-1635S, Super AllWrite SHM-165P6S and LightScribe

SHW-16H5S, Plextor PX-716A and PX-740A, Samsung SH-W162, and the Sony DRU-810A.

Features

Read and write speeds

The read speeds of all the drives were almost identical, with CD-ROM read speeds of 40x and 48x, and DVD-ROM read speeds of 16x. 16x is pretty good; whether you're watching DVD movies or copying files, none of these drives should pose a speed problem.

Similarly, write speeds were almost identical across the brands. The BenQ DW1625 seemed to be left behind, partly because this is one of the older models. It supported writing to DVD-R at a maximum speed of 8x, as against the 16x supported by the rest of the writers. In fact, neither of the BenQ models can be called new; they've been around for a while. That's also the reason why neither of them supports the DVD-R DL standard.

Multi-drives

There were three multi-drives in this test: the Asus DRW-1608P2S, Lite-On Super AllWrite SHM-165P6S, and the LG GSA-4167B. A multi-drive supports reading from and writing to all existing DVD formats, including DVD-RAM. The ability to write to DVD-

How We Tested

All the DVD-Writers were tested on a single test machine to obtain comparable results. The test machine we used comprised an Intel Pentium IV Extreme Edition processor running at a stock speed of 3.4 GHz, a Foxconn 955X7AA-8EKRS2 motherboard, 1 GB of Micron 533 MHz DDR RAM, an ATI Radeon X850 Platinum Edition 256 MB PCIe card, and a Maxtor Maxline III 250 GB SATA150 hard drive. The test machine was powered by an Antec Neo480 480-watt power supply. Windows XP Professional with SP1 along with the latest drivers was installed afresh on the test machine. Nero Burning ROM 7 Ultra Edition was installed for the purpose of burning CDs and DVDs. The internal drives were set as master and connected to the primary IDE channel. The external drives were connected using the USB 2.0 interface.

The test process consisted of two sections—reading and writing of different types of media.

Reading Tests

Nero CD-DVD Speed was used to record data transfer rates, access/seek times, CPU utilisation, burst rate, etc., for CDs as well as DVDs. This software comes bundled with Nero Burning ROM 7.

SiSoft Sandra 2005 SR3, which is a system-wide performance benchmark as well as a diagnostic utility, was used to measure read speeds and access times.

dbPowerAmp, which is an audio format conversion and audio CD ripping tool, was used to rip pre-selected audio tracks of our

test audio CD. The tracks were ripped to the MP3 format at 44 KHz and 128 Kbps, and the time was logged.

DVD Decrypter is a tool used to rip DVD video files to the hard drive. This software was used to rip a pre-selected chapter of our test DVD to the hard drive, and the time taken was noted.

Writing Tests

These included logging the time taken to write to CD-R, CD-RW, DVD-R, DVD+R, DVD-RW and DVD+RW. Assorted data totalling 700 MB was used to write to CD-R, CD-RW and DVD-RW, and the time was logged. Sequential data consisting of a 700 MB file was used to write to CD-R. Similarly, assorted and sequential data of 4.5 GB was used to write to DVD-R and DVD+R media, and the time taken was noted.

The media used were 52x CD-R from EuroVision, 10x CD-RW from Moser Baer, 8x DVD-R from Moser Baer, 8x DVD+R from Sony, and 2x DVD-RW from Moser Baer. A point to note is that though most DVD-Writers can burn DVDs at 16x, this media is not easily available in Indian markets even now, and even if it is, the costs are prohibitive. We therefore used the more commonly available and more affordable 8x media to compare the drives.

Other features such as support for various DVD types, maximum read and write speeds, LightScribe capability, tray build quality, silent operation, and bundled accessories such as the data cable, manual, media, bezels, etc. were noted and rated to arrive at the final score for a drive.